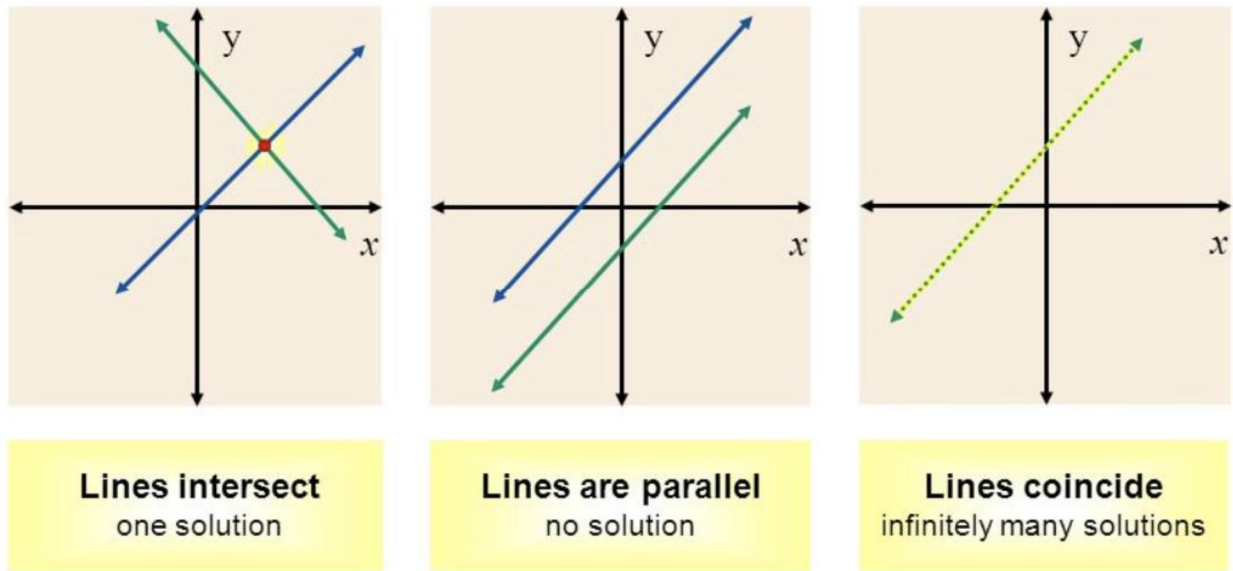


## System of Linear Equation

Relation between two straight lines:



**Interpretation:**

❖ 2 or more straight lines constitute a system.

Relation between two lines,

- If two straight lines intersect each other, then create 1 common point of them.
- If two straight lines are parallel, then there is no common points.
- If one straight lines overlap to another straight lines, then there is infinity common points.

**Some necessary concept:**

- Every equation like  $ax + by + c = 0$  represents a straight line.
- The equation  $ax + by + c = 0$  is a linear equation in 2 variables.
- More than one equations will make a system of linear equations.

The equations of the system may have

- i. Only 1 common solution or unique solution.
- ii. No common solution or no solutions.
- iii. Infinite solutions or many solutions.

## GENERALIZATION:

- The equation

$$a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_nx_n = 0 \dots \dots \dots (1)$$

is a linear equation in  $n$  variables.

- A SLE consists of  $m$  linear equations, each in  $n$  variables. It may be written as follows.

$$\left. \begin{array}{l} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2 \\ \dots \dots \dots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n = b_m \end{array} \right\} \dots \dots \dots (2)$$

- The SLE will be called **homogeneous** if the RHS of each equations are equal to zero.
- The SLE will be called **non-homogeneous** if the RHS of each equations are not equal to zero.

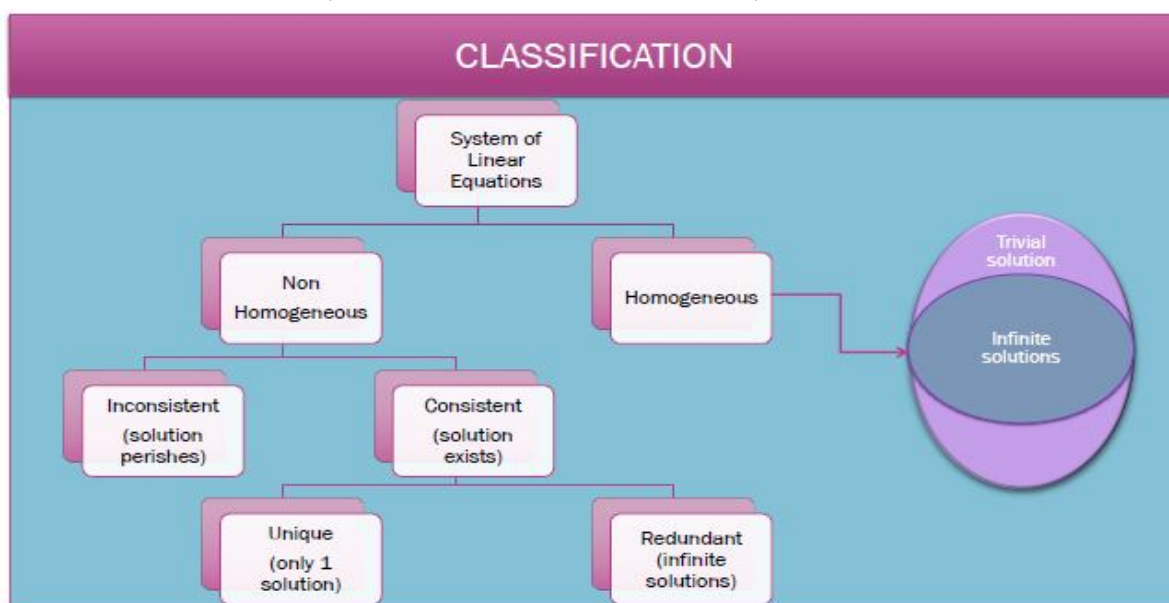
## MATRIX FORM OF SLE:

Matrix Form

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

$$AX = b$$

Where A: coefficient matrix; X: Unknown or Solution Matrix; b: Constant Matrix.



## METHODS FOR SOLUTION:

- Matrix Method
- Determinant Method
- REM/RREM Method

**Example:** Solve the System of Linear Equations,

$$-5a + 3b - 3c = 17$$

$$4a - 3b + 8c = 9$$

$$8a + 5b = -2$$

**Solution:** The given system is converted to matrix form,

$$\begin{bmatrix} -5 & 3 & -3 \\ 4 & -3 & 8 \\ 8 & 5 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 17 \\ 9 \\ -2 \end{bmatrix}$$

$$\Rightarrow AX = b \text{ [Say]}$$

$$\Rightarrow A^{-1}AX = A^{-1}b$$

$$\Rightarrow IX = A^{-1}b$$

$$\Rightarrow X = A^{-1}b$$

$$\Rightarrow \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \frac{1}{260} \begin{bmatrix} -40 & -15 & 15 \\ 64 & 24 & 28 \\ 44 & 49 & 3 \end{bmatrix} \begin{bmatrix} 17 \\ 9 \\ -2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} -13/4 \\ 24/5 \\ 91/20 \end{bmatrix}$$

$$\therefore a = -\frac{13}{4}; b = \frac{24}{5}; c = \frac{91}{20}$$

## Limitations:

1. If M is not square, this method can't be applied.
2. Does not work for the system when  $|M| = 0$ .
3. Becomes too much lengthy when M is 4x4, 5x5, or so on.

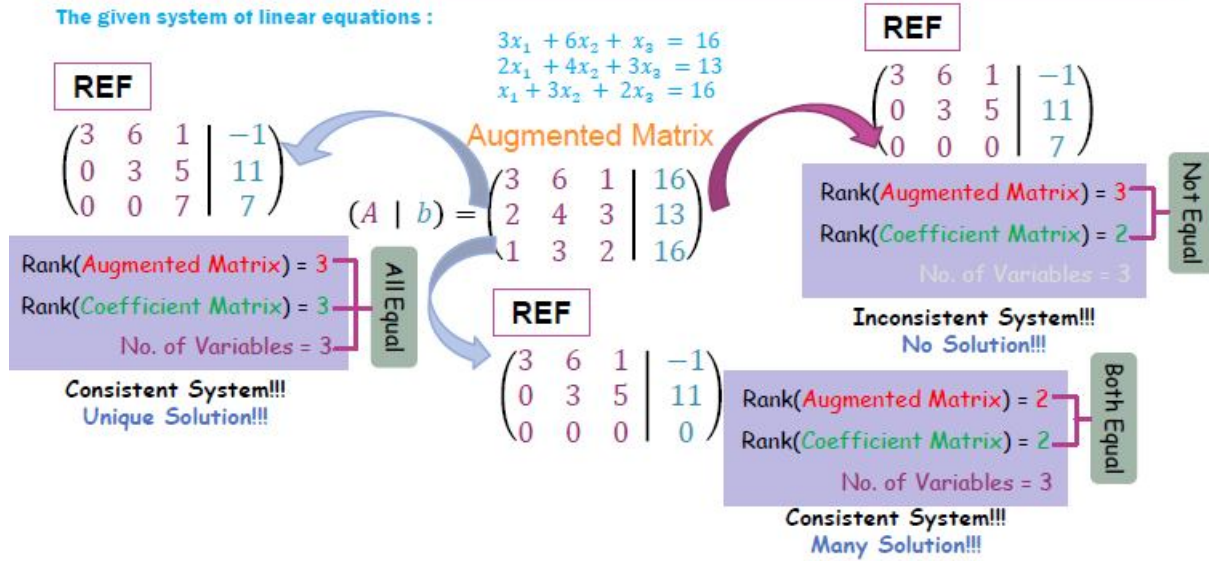
## WHY REM/RREM METHOD?

All problems related to the solution are overcome when we apply the general REM/RREM method, which depends on the REF/RREF method, to solve a SLE.

## SOLUTION OF NON-HOMOGENEOUS SLE

The given system of linear equations :

$$\begin{aligned} 3x_1 + 6x_2 + x_3 &= 16 \\ 2x_1 + 4x_2 + 3x_3 &= 13 \\ x_1 + 3x_2 + 2x_3 &= 16 \end{aligned}$$



## CONSISTENT SLE: UNIQUE SOLUTION

Solve the system of linear equations :

$$\begin{aligned} 3x_1 + 6x_2 + x_3 &= 16 \\ 2x_1 + 4x_2 + 3x_3 &= 13 \\ x_1 + 3x_2 + 2x_3 &= 16 \end{aligned}$$

**Solution:** The augmented matrix of the system is:

$$(A | b) = \left( \begin{array}{ccc|c} 3 & 6 & 1 & 16 \\ 2 & 4 & 3 & 13 \\ 1 & 3 & 2 & 16 \end{array} \right) \sim \left( \begin{array}{ccc|c} 3 & 6 & 1 & 16 \\ 0 & 0 & 7 & 7 \\ 0 & 3 & 5 & 32 \end{array} \right) \begin{array}{l} R_2 \rightarrow 3R_2 - 2R_1 \\ R_3 \rightarrow 3R_3 - R_1 \end{array}$$

$$\sim \left( \begin{array}{ccc|c} 3 & 6 & 1 & 16 \\ 0 & 3 & 5 & 32 \\ 0 & 0 & 7 & 7 \end{array} \right) \begin{array}{l} R_2 \leftrightarrow R_3 \\ \text{REF} \end{array}$$

**Rank(Augmented Matrix) = 3**  
**Rank(Coefficient Matrix) = 3**  
 No. of Variables = 3

**All Equal**

**Consistent System!!!**  
**Unique Solution!!!**

No. of Free Variables = No. of Unknown (Variables) - No. of Pivot = 3 - 3 = 0

Hence the system is **consistent** and therefore the system has **unique solution**.

**By back substitution:**

$$7x_3 = 7 \Rightarrow x_3 = 7/7 = 1$$

$$3x_2 + 5x_3 = 32 \Rightarrow x_2 = \frac{1}{3}(32 - 5x_3) = \frac{1}{3}(32 - 5 \times 1) = 9$$

$$3x_1 + 6x_2 + x_3 = 16 \Rightarrow x_1 = \frac{1}{3}(16 - 6x_2 - x_3) = \frac{1}{3}(16 - 6 \times 9 - 1) = -13$$

## CONSISTENT SLE: MANY SOLUTION

Solve the system of linear equations:

$$\begin{aligned} 2x - 4y + 6z &= 8 \\ 3x + 2y + z &= 4 \\ -x + 2y - 3z &= -4 \end{aligned}$$

$$\text{Rank(Augmented Matrix)} = 2$$

$$\text{Rank(Coefficient Matrix)} = 2$$

$$\text{No. of Variables} = 3$$

Both Equal

**Solution:** The augmented matrix of the system is:

$$(A | b) = \left( \begin{array}{ccc|c} 2 & -4 & 6 & 8 \\ 3 & 2 & 1 & 4 \\ -1 & 2 & -3 & -4 \end{array} \right) \sim \left( \begin{array}{ccc|c} 2 & -4 & 6 & 8 \\ 0 & 16 & -16 & -16 \\ 0 & 0 & 0 & 0 \end{array} \right) \begin{array}{l} R_2 \rightarrow 2R_2 - 3R_1 \\ R_3 \rightarrow 2R_3 + R_1 \end{array}$$

REF

**Consistent System!!!**  
**Many Solution!!!**

$$\text{No. of Free Variables} = \text{No. of Unknown (Variables)} - \text{No. of Pivot} = 3 - 2 = 1$$

Hence the system is **consistent** and therefore the system has **many solution** (redundant solution).

Here the free variable is  $z$  and let  $z = t$  (where  $t$  is arbitrary constant)

**By back substitution: (General Solution)**

$$z = t$$

$$0 \cdot x + 16y - 16z = -16 \Rightarrow y = \frac{1}{16}(-16 + 16z) = (-1 + z) = (-1 + t) = t - 1$$

$$2x - 4y + 6z = 8 \Rightarrow x = \frac{1}{2}(4y - 6z) = \frac{1}{2}[8 + 4(t - 1) - 6t] = \frac{1}{2}(4 - 2t) = 2 - t$$

## CONSISTENT SLE: MANY SOLUTION (CONT...)

Solve the system of linear equations:

$$\begin{aligned} 2x - 4y + 6z &= 8 \\ 3x + 2y + z &= 4 \\ -x + 2y - 3z &= -4 \end{aligned}$$

**Solution(continue):**

**By back substitution: (General Solution)**

$$z = t$$

$$0 \cdot x + 16y - 16z = -16 \Rightarrow y = \frac{1}{16}(-16 + 16z) = (-1 + z) = (-1 + t) = t - 1$$

$$2x - 4y + 6z = 8 \Rightarrow x = \frac{1}{2}[8 + 4(t - 1) - 6t] = \frac{1}{2}(4 - 2t) = 2 - t$$

**General Solutions are:**

$$x = 2 - t, \quad y = t - 1, \quad z = t$$

**Particular Solution:**

For particular solution let  $t = 1$  then

$$x = 1, \quad y = 0, \quad z = 1$$

## INCONSISTENT SLE: NO SOLUTION

Solve the system of linear equations by using gaussian elimination method:

$$\begin{aligned}x_1 + 2x_2 - 3x_3 &= -1 \\5x_1 + 3x_2 - 4x_3 &= 2 \\3x_1 - x_2 + 2x_3 &= 7\end{aligned}$$

**Solution:** The augmented matrix of the system is:

$$\begin{aligned}(A | b) &= \left( \begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 5 & 3 & -4 & 2 \\ 3 & -1 & 2 & 7 \end{array} \right) \\&\sim \left( \begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 0 & -7 & 11 & 7 \\ 0 & -7 & 11 & 10 \end{array} \right) \begin{array}{l} R_2 \rightarrow R_2 - 5R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{array} \\&\sim \left( \begin{array}{ccc|c} 1 & 2 & -3 & -1 \\ 0 & -7 & 11 & 7 \\ 0 & 0 & 0 & 3 \end{array} \right) \begin{array}{l} R_3 \rightarrow R_3 - R_2 \\ \text{REF} \end{array}\end{aligned}$$

$$\begin{aligned}\text{Rank}(\text{Augmented Matrix}) &= 3 \\ \text{Rank}(\text{Coefficient Matrix}) &= 2 \\ \text{No. of Variables} &= 3\end{aligned}$$

Not Equal

**Inconsistent System!!!**  
**No Solution!!!**

The system contains an equation of the form  $0 = 3$  (which is impossible) .

Hence the system is **inconsistent** and therefore the system has **no solution**.

### Exercise

- When does an SLE become consistent?
- How many free variables do exist for a unique solution of SLE?
- An SLE can't have all free variables. True or false?
- Write down the condition of consistency of an SLE?
- Write down the condition of an SLE has a unique solution and when it is redundant.

Solve the following SLE.

$$3x - 4y + 5z = 0$$

$$1. \quad x + y - 7z = 0$$

$$6x + y - z = -1$$

$$7x - y + z + t = 0$$

$$3. \quad x + 9y - z = -8$$

$$2x + 5z - 3t = 4$$

$$x - y - z - t = -5$$

$$x + y + z = 0$$

$$2. \quad -x + y - z = 9$$

$$4x - 7y + 9z = 1$$

$$a + b - c - d = 7$$

$$4. \quad -2a + b - d = 0$$

$$a - b + c - d = 9$$

$$5a + 4b + 3c + 2d = 1$$